

The covalent bonds between arsenic and gallium atoms in a gallium arsenide compound are strong bonds, still it is possible to break the bonds, if sufficient energy is applied externally. Due to the breaking of covalent bonds between arsenic and gallium, electrons come out from the lattice structure of the GaAs compounds. An electron on its separation from the covalent bond leaves a vacancy behind it in the bond. The separated electrons from the bonds are free to move anywhere in the lattice and are referred to as free electrons. Similarly, the vacancies created in the bonds are referred to as holes, and both free electrons and holes are called free charge carriers.

The energy gap between valence band and conduction band in GaAs is 1.43eV, which is the largest band gap as compared to other common semiconductors. Among the three most popular semiconductor materials, the use of GaAs is growing. Because of very high speed operating characteristics, low reverse saturation currents, excellent temperature sensitivities, and high breakdown voltages associated with GaAs wafers, nowadays, the use of GaAs in place silicon has gained ground in manufacturing very large scale integrated circuits. With all these advantages, GaAs is also being widely used in other different optoelectronics applications like LEDs, solar cells, and photodetector devices.

Electron mobility, single junction band-gap, higher efficiency, heat and moisture resistance, and superior flexibility are the five distinct advantages of GaAs, which are improving the acceptance of GaAs wafers in the semiconductor industry. Liquid encapsulation Czochralski (LEC) and vertical gradient freeze (VGF) are two popular methods which are improving the production of GaAs wafers with high uniformity of electrical properties and excellent surface quality.

Unique applications and opportunities

The performance of high-speed semiconductor devices, which almost drive the present-day digital computers, electronic systems for digital signal processing, telecommunica-

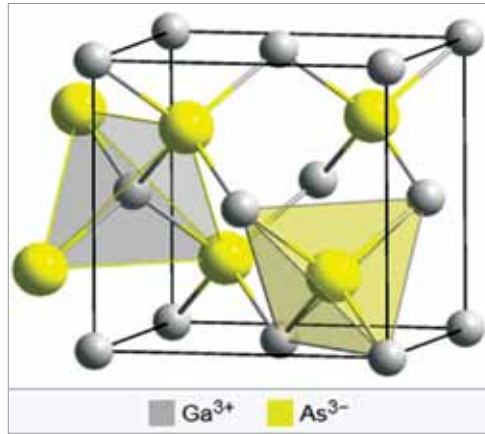


Fig. 2: Unit cell structure of GaAs

tion systems, and optoelectronics are undistinguishably linked to the unique physical and electrical properties offered by GaAs. Once viewed as a simple alternative to silicon, gallium arsenide has emerged as an irreplaceable material in component fabrication in leading high-tech industries.

Moreover, the increasing expertise with the fabrication methods, operating principles, device models, novel device designs, the material properties, and physics of GaAs is so integral to this success. GaAs high-speed devices provide a comprehensive and state-of-the-science look. Gallium arsenide eventually has emerged as the favourite material for high-frequency applications due to the high carrier mobility compared with silicon devices.

Wide energy band gap makes pure GaAs highly resistive and, combined with a high dielectric constant, this property makes GaAs a very good substrate for integrated circuits. This provides natural isolation between devices and circuits with GaAs, unlike silicon. This advantage has made it an ideal material in monolithic microwave integrated circuits (MMICs). GaAs devices are particularly useful in very high speed computer circuits and at microwave frequencies above 4GHz in discrete components or microwave monolithic integrated circuits (MMICs).

GaAs based microwave devices have found extensive use in consumer applications as well in sophisticated warfare electronics. The use of GaAs in transistors and integrated circuits

for space and military applications has greatly expanded over the past few years.

The unique property of gallium arsenide is its ability to turn electricity into light. Thus, GaAs also exhibits relevant optical properties and finds extensive applications in optoelectronics including LEDs, laser diodes, solar cells, optical windows, and GaAs diodes to detect X-rays. Several solar car models are coming that make use of GaAs in solar arrays.

The main reasons for the exploitation of this semiconductor are that GaAs devices can operate at higher frequencies and have greater radiation hardness than their silicon counterparts. GaAs applications are growing in different fields and it is being used in the manufacture of devices such as microwave frequency and monolithic microwave integrated circuits, infrared LEDs, laser diodes, solar cells, and optical windows. GaAs is also often used as a substrate material for the epitaxial growth of other III-V group semiconductors like indium gallium arsenide and aluminium gallium arsenide. GaAs has been used for various transistor types including the following:

- Metal semiconductor field-effect transistor (MESFET)
- High-electron-mobility transistor (HEMT)
- Junction field-effect transistor (JFET)
- Heterojunction bipolar transistor (HBT)
- Metal oxide semiconductor field-effect transistor (MOSFET)

The superior properties of GaAs compared to silicon, such as higher saturated electron velocity and higher electron mobility, allow GaAs transistors to function at higher frequencies. Due to its wider energy band gap, GaAs devices are relatively insensitive to overheating. As a result of higher carrier motilities and lower resistive device parasitics, they also tend to create less noise (disturbance in an electrical signal) in electronic circuits than silicon devices, especially at high frequencies. These superior properties are compelling scientists, engineers, and manufacturers to think about